

TCS - 100 Technical MCQ Questions

Tata Consultancy Services

For BTech Computer Science / IT Students

Object-Oriented Programming

- 1. An abstract class in Java can have:**
 - A) Only abstract methods
 - B) Only concrete methods
 - C) Both abstract and concrete methods
 - D) No methods at all
- 2. Which of the following is true about a constructor?**
 - A) It has a return type
 - B) It is called automatically when an object is created
 - C) It can be inherited like other methods
 - D) It must be private
- 3. Which keyword is used in Java to prevent a class from being inherited?**
 - A) static
 - B) final
 - C) private
 - D) abstract
- 4. Which of these is an example of compile-time polymorphism?**
 - A) Method overriding
 - B) Operator overloading (in C++)
 - C) Dynamic method dispatch
 - D) Runtime type identification
- 5. What does 'this' keyword refer to in Java/C++?**
 - A) The parent class
 - B) The current object instance
 - C) A static variable
 - D) The compiler
- 6. Which of the following supports multiple inheritance in Java?**
 - A) Classes
 - B) Interfaces
 - C) Constructors
 - D) Packages
- 7. Which of these best defines an interface in Java?**
 - A) A class with only static methods
 - B) A contract of methods a class must implement
 - C) A private class
 - D) A type of exception

8. Which access modifier makes a member accessible only within the same class?

- A) public
- B) protected
- C) private
- D) default

9. Constructor overloading is possible when:

- A) Constructors have same name but different parameter lists
- B) Constructors are private
- C) A class has no constructor
- D) Only one constructor exists

10. What is the purpose of the 'super' keyword in Java?

- A) To create a static method
- B) To refer to the parent class members
- C) To delete an object
- D) To define an interface

11. A class that cannot be instantiated directly is called a(n):

- A) Abstract class
- B) Final class
- C) Static class
- D) Sealed class

12. What is encapsulation?

- A) Hiding implementation details and exposing only functionality
- B) Creating multiple classes
- C) Compiling code faster
- D) Using loops effectively

13. Which concept allows objects of different classes to be treated as objects of a common superclass?

- A) Polymorphism
- B) Encapsulation
- C) Abstraction
- D) Overloading

14. What happens if a class does not implement all abstract methods of an interface it implements?

- A) Compilation succeeds silently
- B) The class must be declared abstract
- C) Runtime error occurs
- D) Nothing happens

15. What is a virtual function used for in C++?

- A) To achieve runtime polymorphism
- B) To avoid inheritance
- C) To create static memory
- D) To improve compilation speed

16. What is the main difference between an abstract class and an interface (pre-Java 8)?

- A) Interfaces can have constructors
- B) Abstract classes cannot have any methods
- C) Interfaces cannot have method bodies while abstract classes can
- D) There is no difference

DBMS & SQL

17. What is a primary key?

- A) A column that can contain duplicate values
- B) A column (or set of columns) that uniquely identifies each row and cannot be NULL
- C) An index used only for sorting
- D) A key used only in foreign tables

18. Which of the following best defines a stored procedure?

- A) A precompiled collection of SQL statements stored in the database and executed as a unit
- B) A type of foreign key
- C) A column constraint
- D) A backup mechanism

19. What is a view in SQL?

- A) A physical copy of a table
- B) A virtual table based on the result of a SQL query
- C) A type of index
- D) A stored procedure

20. What is denormalization?

- A) The process of adding redundancy to improve read performance
- B) Removing all keys from a table
- C) A type of join
- D) Encrypting a database

21. Which SQL statement is used to combine the result-set of two or more SELECT statements with the same number of columns?

- A) JOIN
- B) UNION
- C) MERGE
- D) GROUP BY

22. What does ACID stand for in DBMS transactions?

- A) Atomicity, Consistency, Isolation, Durability
- B) Accuracy, Consistency, Integration, Data
- C) Atomic, Complete, Independent, Durable
- D) Association, Consistency, Isolation, Data

23. Which SQL command is used to change the structure of an existing table?

- A) UPDATE
- B) ALTER
- C) MODIFY
- D) CHANGE

24. What does the term 'referential integrity' ensure?

- A) That foreign key values always match a primary key value in the referenced table (or are NULL)
- B) That all columns are indexed
- C) That all queries run faster
- D) That tables cannot be dropped

25. Which property of a transaction ensures that partial updates are never visible to other transactions?

- A) Atomicity
- B) Consistency
- C) Isolation
- D) Durability

26. Which of the following commands removes all rows from a table but keeps its structure, and cannot be rolled back in most databases?

- A) DELETE
- B) DROP
- C) TRUNCATE
- D) REMOVE

27. Which index type stores data rows in the same order as the index?

- A) Non-clustered index
- B) Clustered index
- C) Bitmap index
- D) Hash index

28. Which type of JOIN returns all rows from the left table and matched rows from the right table?

- A) INNER JOIN
- B) LEFT JOIN
- C) RIGHT JOIN
- D) CROSS JOIN

29. What is the difference between WHERE and HAVING clauses?

- A) WHERE filters rows before grouping, HAVING filters groups after aggregation
- B) They are identical in function
- C) HAVING is used only with SELECT *
- D) WHERE cannot use comparison operators

30. Which SQL aggregate function returns the number of rows matching a condition?

- A) SUM()
- B) COUNT()
- C) AVG()
- D) MAX()

31. What is a deadlock in the context of a database transaction?

- A) Two or more transactions waiting indefinitely for resources locked by each other
- B) A syntax error in SQL
- C) A missing index
- D) A type of join

32. What is the purpose of an index in a database?

- A) To enforce referential integrity
- B) To speed up data retrieval operations
- C) To store backup data
- D) To encrypt table data

33. Which normal form eliminates transitive dependency?

- A) 1NF
- B) 2NF
- C) 3NF
- D) 0NF

34. Which of the following is true about a candidate key?

- A) It can contain NULL values
- B) Every candidate key can be a primary key, but only one is chosen as such
- C) A table can have only one candidate key
- D) It must always be a foreign key

35. What is a composite key?

- A) A key formed by combining two or more columns to uniquely identify a row
- B) A key that references another table
- C) An automatically generated key
- D) A key with only one column

36. Which SQL clause is used to filter groups after a GROUP BY?

- A) WHERE
- B) HAVING
- C) FILTER
- D) ORDER BY

Operating Systems

37. Which page replacement algorithm suffers from Belady's Anomaly?

- A) LRU
- B) FIFO
- C) Optimal
- D) LFU

38. What is the purpose of a context switch?

- A) To save the state of a running process and load the state of another
- B) To terminate a process
- C) To allocate memory
- D) To format a disk

39. Which of the following synchronization tools is used to prevent race conditions?

- A) Compiler
- B) Semaphore
- C) Scheduler
- D) Loader

40. What is the purpose of a semaphore?

- A) To manage access to shared resources and avoid race conditions
- B) To compile code
- C) To manage file names
- D) To render graphics

41. What is virtual memory?

- A) Physical RAM only
- B) A memory management technique that gives an application the illusion of a large, contiguous memory using disk space
- C) A type of cache memory
- D) A backup memory device

42. What is the difference between a process and a thread?

- A) A process is a lightweight version of a thread
- B) A thread is an independent execution unit within a process, sharing the process's memory space
- C) Threads have separate memory spaces from their process
- D) There is no difference

43. Which memory management technique divides memory into fixed-size blocks?

- A) Segmentation
- B) Paging
- C) Swapping
- D) Caching

44. Which scheduling algorithm gives the shortest average waiting time (in theory, for a given batch)?

- A) FCFS
- B) SJF (Shortest Job First)
- C) Round Robin with a large quantum
- D) Priority Scheduling

45. Which of the following is an example of an inter-process communication (IPC) mechanism?

- A) Pipes
- B) Compilers
- C) Loaders
- D) Linkers

46. Which of these describes a 'zombie process'?

- A) A process that has completed execution but still has an entry in the process table
- B) A process that never starts
- C) A process using 100% CPU
- D) A kernel-level thread

47. What does 'thrashing' refer to in an OS?

- A) Excessive CPU usage by a single process
- B) A state where the system spends most of its time swapping pages instead of executing processes
- C) A type of deadlock
- D) A network congestion issue

48. What is a race condition?

- A) When multiple processes/threads access shared data concurrently and the outcome depends on timing
- B) A process running faster than expected
- C) A type of scheduling algorithm
- D) A network protocol

49. Which of the following is a preemptive scheduling algorithm?

- A) FCFS
- B) Round Robin
- C) Non-preemptive SJF
- D) Batch scheduling

50. Which of the following is NOT one of the four necessary conditions for deadlock?

- A) Mutual Exclusion
- B) Hold and Wait
- C) Preemption
- D) Circular Wait

51. What is the difference between paging and segmentation?

- A) Paging divides memory into fixed-size blocks; segmentation divides memory into variable-sized logical units
- B) They are identical
- C) Segmentation is faster in all cases
- D) Paging is used only for disk storage

52. What is a critical section in OS?

- A) A section of code that accesses shared resources and must not be executed by more than one process/thread at a time
- B) A section reserved for the kernel only
- C) A type of page fault
- D) A memory segment for stack only

Computer Networks

53. Which of the following best describes a subnet mask?

- A) It divides an IP address into network and host portions
- B) It encrypts network traffic
- C) It assigns MAC addresses
- D) It is used only in IPv6

54. Which protocol is used to resolve a domain name to an IP address?

- A) FTP
- B) DNS
- C) SMTP
- D) ARP

55. What is NAT (Network Address Translation) used for?

- A) To translate private IP addresses to a public IP address for internet access
- B) To encrypt data
- C) To assign MAC addresses
- D) To resolve domain names

56. Which protocol operates at the Transport layer and is connection-oriented?

- A) UDP
- B) IP
- C) TCP
- D) ICMP

57. What is a firewall used for?

- A) To speed up internet connections
- B) To monitor and control incoming/outgoing network traffic based on security rules
- C) To assign IP addresses
- D) To translate domain names

58. Which protocol is used to send emails?

- A) POP3
- B) SMTP
- C) HTTP
- D) FTP

59. Which of the following devices operates primarily at the Network layer to forward packets between different networks?

- A) Switch
- B) Hub
- C) Router
- D) Repeater

60. Which layer of the OSI model handles encryption and data formatting?

- A) Presentation Layer
- B) Session Layer
- C) Application Layer
- D) Physical Layer

61. What does DHCP stand for?

- A) Dynamic Host Configuration Protocol
- B) Data Host Control Protocol
- C) Direct Host Connection Protocol
- D) Domain Host Configuration Protocol

62. What is the default port number for HTTPS?

- A) 80
- B) 21
- C) 443
- D) 22

63. What is the default port number for HTTP?

- A) 21
- B) 25
- C) 80
- D) 443

64. What does 'three-way handshake' refer to in TCP?

- A) The SYN, SYN-ACK, ACK process used to establish a connection
- B) A method to terminate a connection
- C) An error-detection mechanism
- D) A DNS lookup process

65. What is the maximum number of usable host addresses in a /24 subnet?

- A) 256
- B) 254
- C) 255
- D) 512

66. What is the purpose of the ARP protocol?

- A) To resolve an IP address to a MAC address
- B) To resolve a domain name to an IP address
- C) To transfer files
- D) To send email

Data Structures & Algorithms

67. Which data structure is typically used to implement BFS?

- A) Stack
- B) Queue
- C) Heap
- D) Array only

68. Which sorting algorithm is considered stable and has $O(n \log n)$ time complexity in all cases?

- A) QuickSort
- B) Merge Sort
- C) Selection Sort
- D) Heap Sort

69. Which data structure is typically used to implement DFS (iteratively)?

- A) Queue
- B) Stack
- C) Heap
- D) Hash table

70. Which traversal of a Binary Search Tree gives elements in sorted order?

- A) Pre-order
- B) Post-order
- C) In-order
- D) Level-order

71. What is the time complexity of accessing an element in an array by index?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n \log n)$

72. What is a balanced binary search tree used to prevent?

- A) Overflow errors
- B) Degeneration into a linked list (worst-case $O(n)$ operations)
- C) Memory leaks
- D) Integer overflow

73. Which of the following problems is a classic example solved using dynamic programming?

- A) Binary search
- B) 0/1 Knapsack problem
- C) Bubble sort
- D) Linear search

74. What is the time complexity of Dijkstra's algorithm using a binary heap?

- A) $O(V^2)$
- B) $O(E \log V)$
- C) $O(V + E)$
- D) $O(E^2)$

75. What is the space complexity of an iterative approach to computing Fibonacci numbers with memoization storing all n results?

- A) $O(1)$
- B) $O(n)$
- C) $O(n^2)$
- D) $O(\log n)$

76. What is the time complexity of inserting an element at the head of a singly linked list?

- A) $O(n)$
- B) $O(1)$
- C) $O(\log n)$
- D) $O(n \log n)$

77. Which data structure uses LIFO (Last In First Out) order?

- A) Queue
- B) Stack
- C) Linked List
- D) Array

78. What is a hash collision?

- A) When two different keys map to the same hash bucket/index
- B) When a hash table runs out of memory
- C) When a key is deleted incorrectly
- D) When two hash tables merge

79. Which data structure is best suited for implementing a priority queue efficiently?

- A) Array
- B) Linked List
- C) Heap
- D) Stack

80. Which algorithmic technique does Merge Sort primarily use?

- A) Greedy
- B) Divide and Conquer
- C) Dynamic Programming
- D) Backtracking

81. What is dynamic programming primarily used for?

- A) Solving problems by breaking them into overlapping subproblems and storing results to avoid recomputation
- B) Sorting arrays only
- C) Random number generation
- D) Compiling code faster

82. What is the average time complexity of searching in a well-implemented hash table?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n^2)$

83. Which of the following is a common technique to resolve hash collisions?

- A) Chaining or open addressing
- B) Binary search
- C) Recursion
- D) Memoization

84. What does BFS stand for in graph traversal?

- A) Best First Search
- B) Breadth First Search
- C) Binary First Search
- D) Backtrack First Search

Programming Language Fundamentals

85. What is a memory leak?

- A) When allocated memory is never freed even though it is no longer needed, wasting resources over time
- B) When a program uses too little memory
- C) A type of compilation error
- D) A CPU scheduling issue

86. What does 'pass by reference' mean?

- A) A copy of the variable's value is passed to the function
- B) The function receives a reference to the actual variable, so changes affect the original
- C) Variables cannot be passed to functions
- D) Only primitive types can be passed

87. In Java, what is the difference between '==' and '.equals()' when comparing objects?

- A) They always behave identically
- B) '==' compares references while '.equals()' compares content (if overridden)
- C) '.equals()' is only for primitives
- D) '==' is only used for Strings

88. Which keyword is used to handle exceptions in Python?

- A) catch
- B) except
- C) rescue
- D) handle

89. What is the purpose of the 'finally' block in exception handling (Java/Python)?

- A) It executes only if an exception occurs
- B) It executes only if no exception occurs
- C) It always executes regardless of whether an exception occurred or not
- D) It is used to declare exceptions

90. What is the difference between a compiler and an interpreter?

- A) A compiler translates the whole program to machine code before execution, an interpreter executes code line-by-line
- B) They are identical
- C) Interpreters are always faster than compilers
- D) Compilers cannot detect syntax errors

91. What is a lambda function/expression?

- A) A named function stored in a file
- B) An anonymous, inline function typically used for short operations
- C) A type of class
- D) A compiler directive

92. In C++, what does the 'const' keyword do when applied to a variable?

- A) Makes the variable a pointer
- B) Prevents the variable's value from being modified after initialization
- C) Declares a static variable
- D) Declares a global variable

93. Which of the following best describes garbage collection?

- A) A manual process where the developer frees memory explicitly
- B) An automatic process that reclaims memory occupied by objects no longer in use
- C) A compilation step
- D) A type of exception handling

94. In Python, which of the following is a mutable data type?

- A) Tuple
- B) String
- C) List
- D) int

Web & Cloud Fundamentals

95. What is the purpose of cookies in web applications?

- A) To store small pieces of data on the client side to maintain state/session information
- B) To encrypt HTTPS traffic
- C) To speed up DNS resolution
- D) To compress images

96. What does CSS stand for?

- A) Cascading Style Sheets
- B) Computer Style Sheets
- C) Creative Style System
- D) Cascading Software Sheets

97. What is auto-scaling in cloud computing?

- A) Automatically adjusting compute resources up or down based on demand
- B) Manually adding servers only
- C) A type of database indexing
- D) A billing model only

98. What is the difference between IaaS, PaaS, and SaaS?

- A) They are all the same cloud service model
- B) IaaS provides raw infrastructure, PaaS provides a platform to build apps, SaaS delivers ready-to-use software
- C) SaaS provides infrastructure while IaaS provides software
- D) PaaS is only for storage

99. Which HTTP method is typically used to retrieve data without side effects?

- A) POST
- B) GET
- C) DELETE
- D) PUT

100. What is the purpose of a Content Delivery Network (CDN)?

- A) To cache and serve content from servers geographically closer to users, reducing latency
- B) To encrypt all web traffic
- C) To replace DNS servers
- D) To manage database transactions

Answer Key

1-C	2-B	3-B	4-B	5-B	6-B	7-B	8-C	9-A	10-B
11-A	12-A	13-A	14-B	15-A	16-C	17-B	18-A	19-B	20-A
21-B	22-A	23-B	24-A	25-C	26-C	27-B	28-B	29-A	30-B
31-A	32-B	33-C	34-B	35-A	36-B	37-B	38-A	39-B	40-A
41-B	42-B	43-B	44-B	45-A	46-A	47-B	48-A	49-B	50-C
51-A	52-A	53-A	54-B	55-A	56-C	57-B	58-B	59-C	60-A
61-A	62-C	63-C	64-A	65-B	66-A	67-B	68-B	69-B	70-C
71-C	72-B	73-B	74-B	75-B	76-B	77-B	78-A	79-C	80-B
81-A	82-C	83-A	84-B	85-A	86-B	87-B	88-B	89-C	90-A
91-B	92-B	93-B	94-C	95-A	96-A	97-A	98-B	99-B	100-A

Format: [Question No.]-[Correct Option Letter]. Content compiled from publicly available 2026 placement-prep resources and established CS-fundamentals question banks - verify current-year specifics before your interview.